

CAPSTONE PROJECT

CUSTOMER BEHAVIOUR ANALYSIS

Presented By

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Problem Statement:

A leading Retail company face challenges in understanding customer shopping behaviour due to the large amount of data generated from different purchasing channels. Changes in customer preferences, product demand, and purchasing patterns make it difficult for businesses to identify the factors that influence consumer decisions and sales performance. Additionally, variations in demographics, product categories, and sales channels (online and offline) create complexity in analyzing customer behaviour. Factors such as discounts, product reviews, seasonal trends, and payment preferences may significantly impact purchasing decisions and repeat purchases, but these relationships are not clearly understood.

Proposed Solution:

1.Data Collection

Obtain the customer behaviour dataset containing information such as customer demographics, product categories, purchase amounts, sales channels, payment methods, discounts, and review ratings.

2.Data Preprocessing

Clean and prepare the dataset by handling missing values, correcting data types, removing duplicates, and transforming variables when required using Python. This step ensures that the data is accurate and suitable for further analysis.

3.Exploratory Data Analysis

Analyze the dataset to identify patterns and relationships in customer purchasing behaviour using MySQL. This includes examining trends across demographics, product categories, seasonal purchases, and online vs offline sales channels.

4.Insight Identification

Identify key factors influencing customer purchasing decisions, such as discounts, review ratings, payment preferences, and seasonal demand. These insights help understand what drives sales and repeat purchases.

5.Data Visualization and Dashboard Creation

Create interactive dashboards using Microsoft Power BI to visualize insights through charts, graphs, and key performance indicators, enabling clear interpretation of customer behaviour and supporting data-driven decision making.

System Approach:

System Requirements

- Computer system with basic processing capability
- Dataset containing customer behaviour and purchase information
- Data analytics tools for preprocessing and analysis
- Microsoft Power BI for dashboard creation and visualization

Libraries / Tools Required

- Python – for data preprocessing and analysis
- Pandas – for data cleaning and manipulation
- NumPy – for numerical operations
- Matplotlib – for basic data visualization
- Seaborn – for statistical data visualization
- MySQL – for data analysis
- Power BI – for interactive dashboard development

Algorithm & Deployment:

Algorithm Selection:

Descriptive and exploratory data analysis techniques are used to study customer purchasing behaviour and identify patterns in the dataset. These methods help understand the relationships between demographics, product categories, and purchasing trends.

Data Input:

The analysis uses features such as customer age, gender, product category, purchase amount, payment method, discount usage, and sales channel (online/offline). These inputs help in understanding factors influencing customer purchasing decisions.

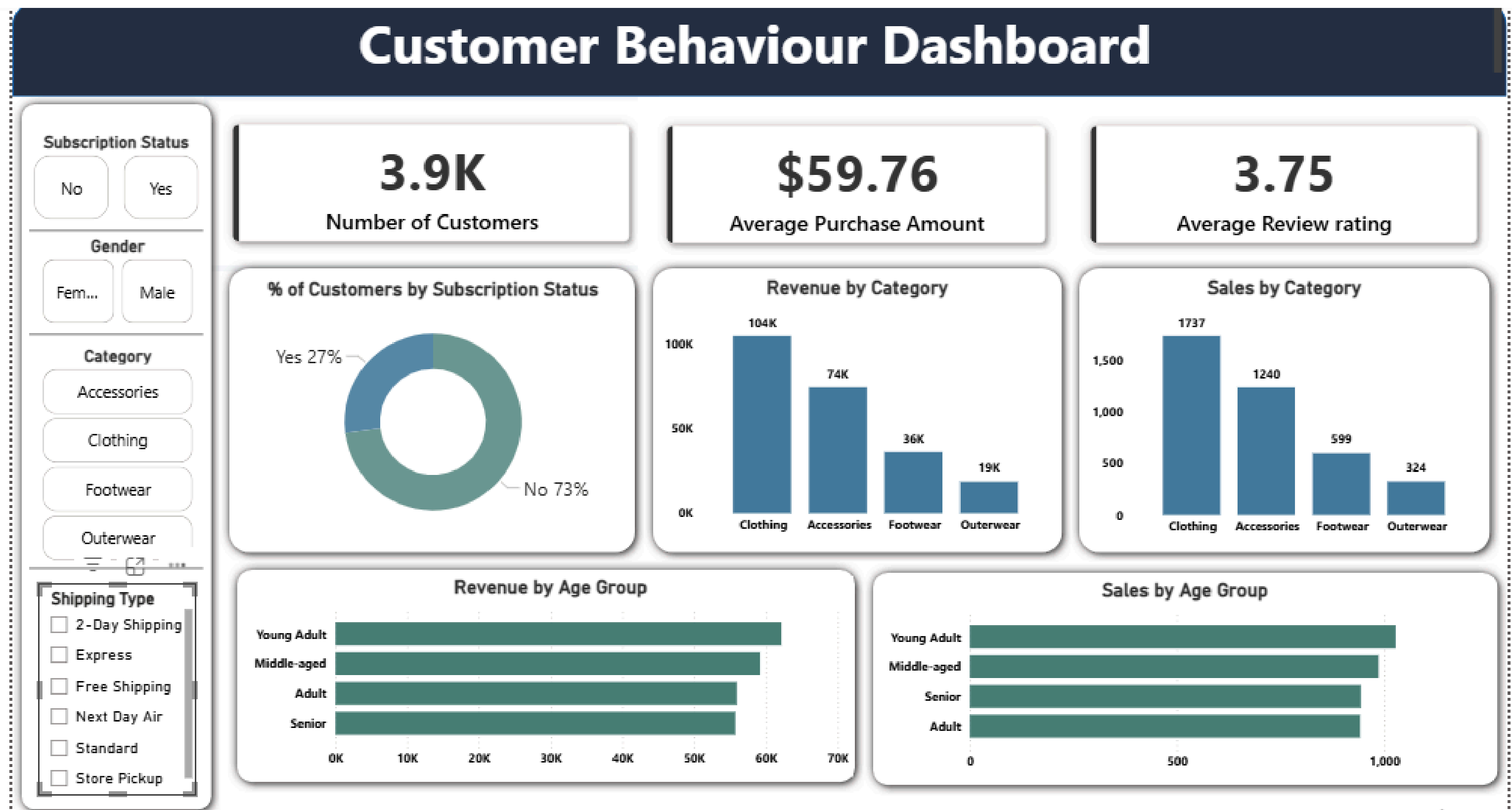
Analysis Process:

The dataset is preprocessed by handling missing values, correcting data types, and cleaning inconsistent data. Exploratory data analysis is then performed to identify trends and relationships within the data.

Insight Generation and Visualization:

The processed data is visualized through interactive dashboards created using Microsoft Power BI. These dashboards present insights that help businesses understand customer behaviour and support data-driven decisions.

Result:



Customer Behaviour Dashboard

Subscription Status

No

Yes

Gender

Fem...

Male

Category

Accessories

Clothing

Footwear

Outerwear

Shipping Type

☐ 2-Day Shipping

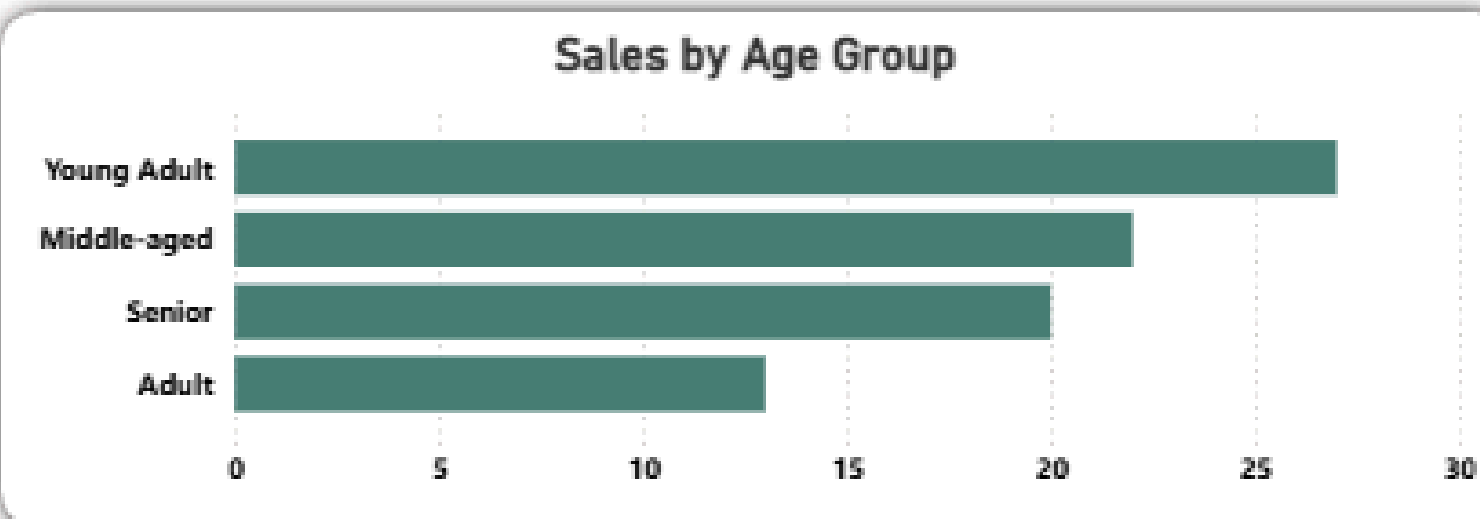
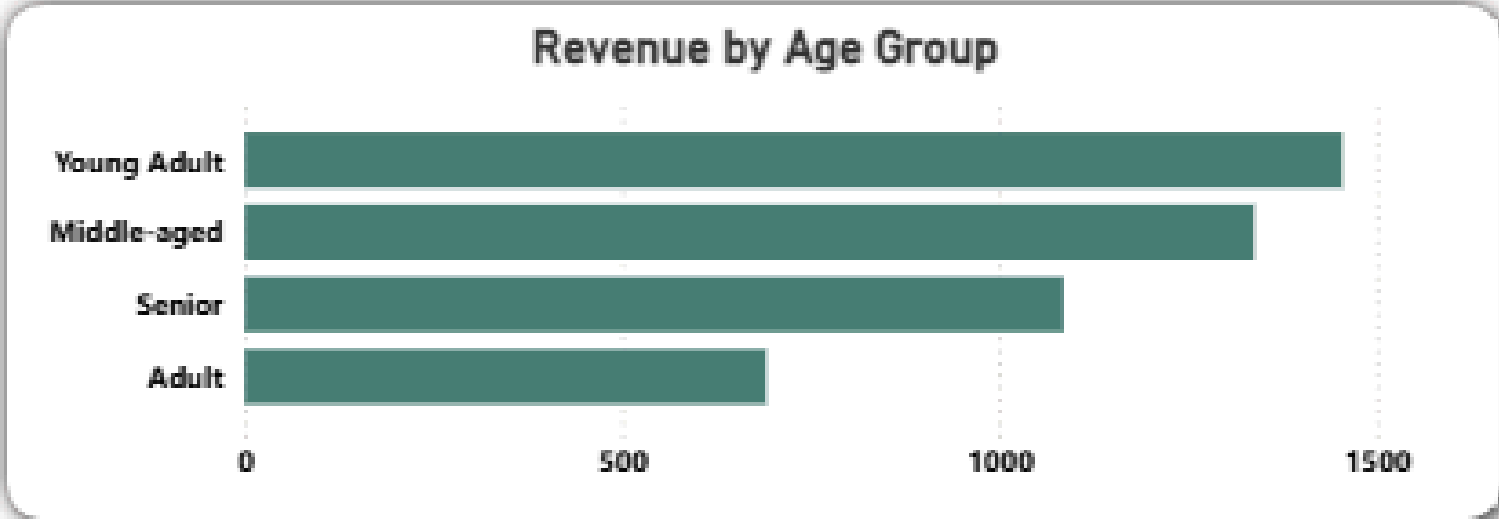
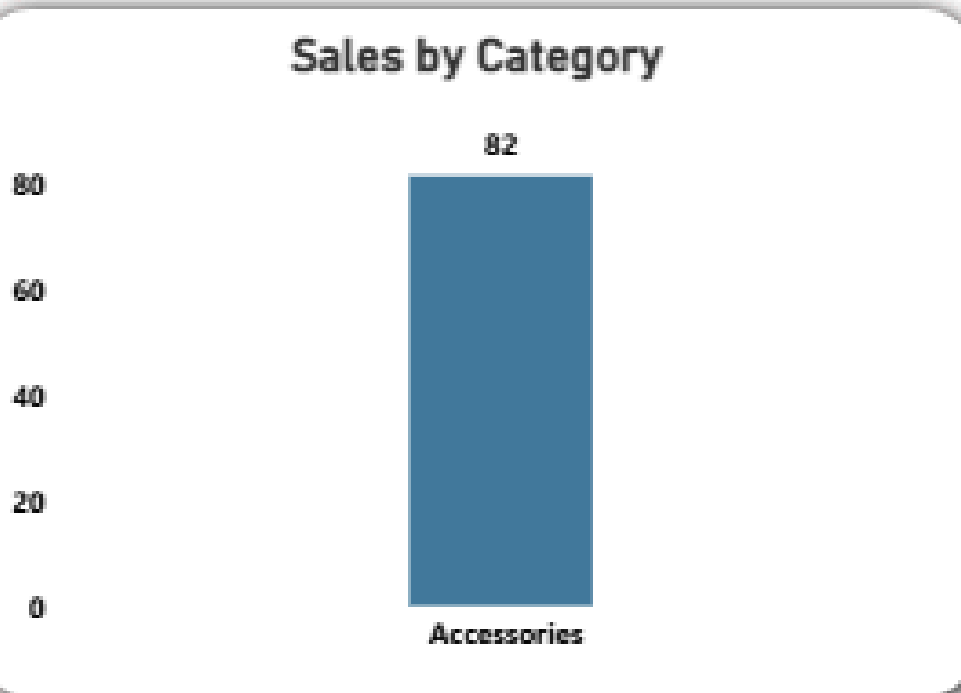
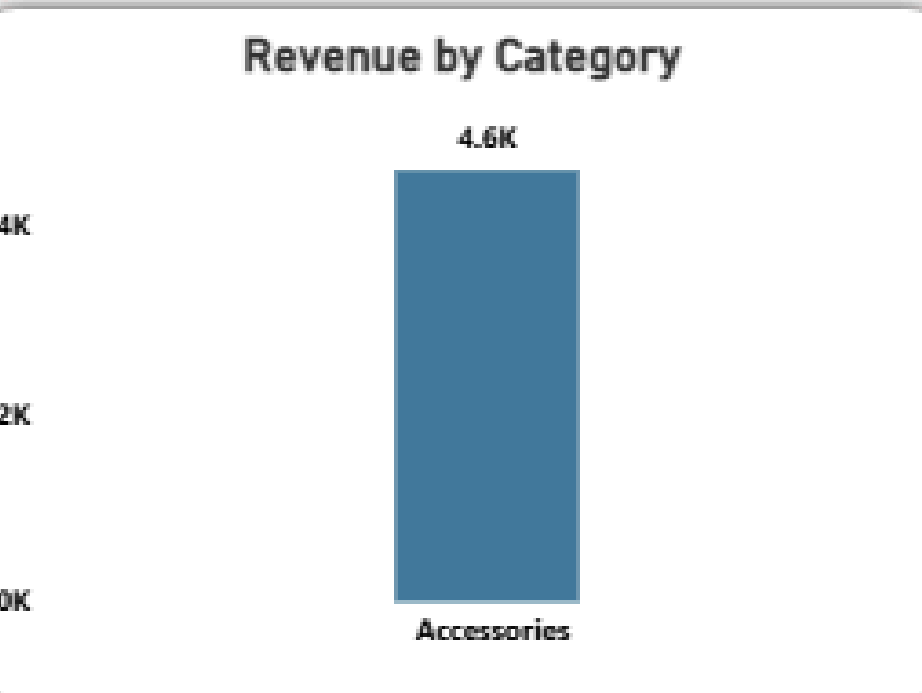
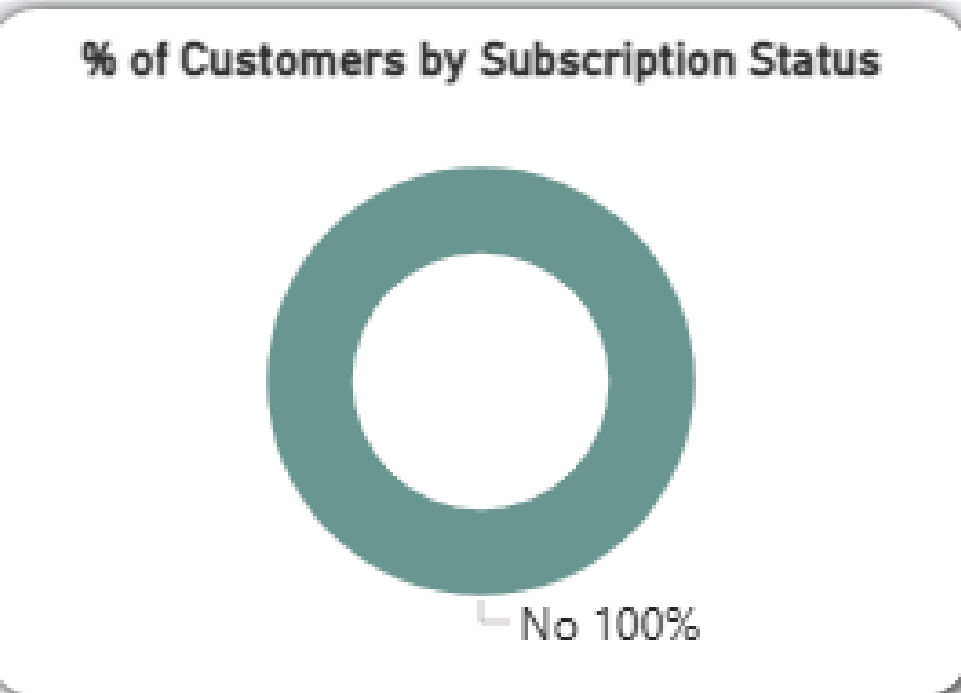
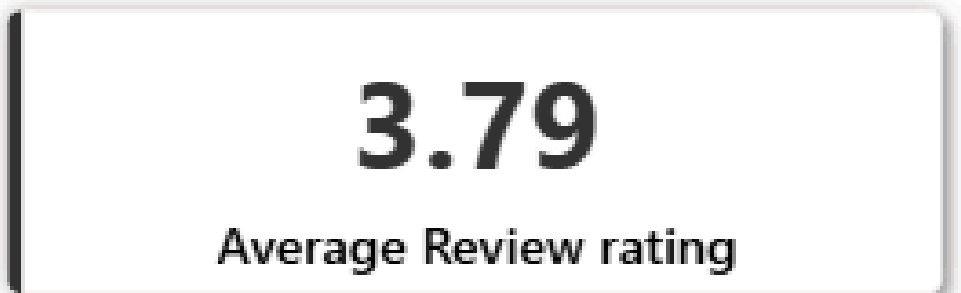
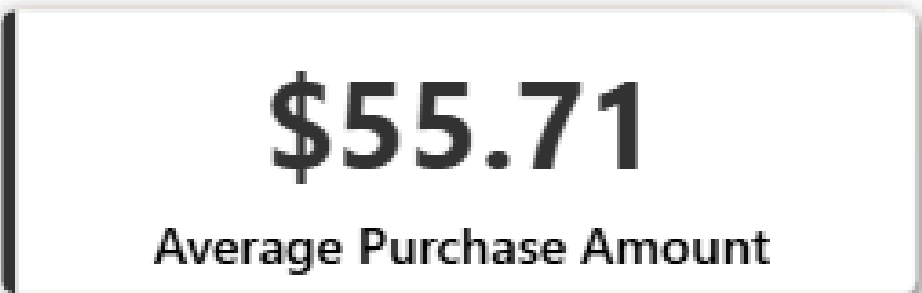
☐ Express

☐ Free Shipping

☐ Next Day Air

☒ Standard

☐ Store Pickup



Conclusion:

Insight 1: Customer Demographics & Revenue Contribution

Male customers generate a higher share of revenue. Young customers contribute the majority of purchases, indicating strong engagement from this segment while other age groups show comparatively lower participation.

gender	revenue
Male	157890
Female	75191

age_group	total_revenue
Young Adult	62143
Middle-aged	59197
Adult	55978
Senior	55763

Insight 2: Product Performance & Customer Preferences

The average product rating is around 4, showing generally positive customer satisfaction. The top-performing products include gloves, sandals, boots, hats, and skirts, and purchase behavior indicates that customers are more likely to buy products when discounts are offered.

item_purchased	discount_perc
Hat	50.00
Sneakers	49.66
Coat	49.07
Sweater	48.17
Pants	47.37

item_purchased	Average Rating
Gloves	3.86
Sandals	3.84
Boots	3.82
Hat	3.8
Skirt	3.78

Future scope:

The customer behaviour analysis system can be enhanced by integrating additional data sources such as customer feedback, website interaction data, and social media insights. Including these sources can provide a deeper understanding of customer preferences and purchasing behaviour. The system can also be expanded to analyze data from multiple stores, regions, or time periods, enabling businesses to gain broader insights into customer trends and market demand. Additionally, advanced analytics and machine learning techniques can be incorporated to predict customer purchasing patterns, identify potential repeat buyers, and detect customer churn. Implementing real-time data processing and automated analytics can further improve the system by providing instant insights and helping businesses make faster, data-driven decisions to improve marketing strategies and customer engagement.

References:

GitHub Link: https://github.com/Saimeghana-guduru/customer_behaviour_analysis.git

Thank You